

Here Comes the Homology

Lecture 5 - CMSE 890

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Tues, Sep 9, 2025

Goals

Goals for today:

- Ch 2.4: Cycles and Boundaries

Section 1

Linear Algebra Review & Homology

Definition

A field $(k, +, \cdot)$ is a set k with two operations $+$ and $\cdot$ such that for any $a, b, c \in k$:

- Closure:
 - ▶
 - ▶
- Commutativity:
 - ▶
 - ▶
- Associativity:
 - ▶
 - ▶
- Identity:
 - ▶
 - ▶

- Inverses:
 - ▶
 - ▶

- Distributivity:

Examples:

Vector space

Definition

A vector space over a field k is a set V with vector addition and scalar multiplication such that

- Associative $+$:
- Commutative $+$:
- Identity

$+$

.

- Inverses:

$+$

- Scalar vs. field mult:

- Distributivity:



Examples:

Definition

A basis for a vector space V is a collection of vectors $\{b_\alpha\}_{\alpha \in A}$ such that

- they are linearly independent and,
- they span V .

Goal

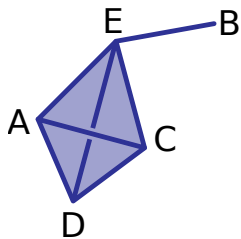
Build a vector space from a simplicial complex!

p -Chains

Let K be a simplicial complex and fix a dimension p .

A p -chain is a finite formal sum of p -simplices in K , written

$$\alpha = \sum a_i \sigma_i$$



Addition of chains

p -chains are added component-wise: if $\alpha = \sum a_i \sigma_i$ and $\beta = \sum b_i \sigma_i$, then

$$\alpha + \beta =$$

Chain group

The collection of p -chains with addition is called the p^{th} -chain group (vector space), $C_p(K)$.

Some checks

- Associative $+$:
- Commutative $+$:
- Zero element
- Inverses:

Linear Transformations

A linear transformation between two vector spaces V and W is a map $T : V \rightarrow W$ such that the following hold:

-
-

Matrix representation:

Boundary maps¹

$$\begin{array}{ccc} \partial_p : & C_p(K) & \rightarrow C_{p-1}(K) \\ & \sigma = [v_0, \dots, v_p] & \mapsto \sum_{j=0}^p [v_0, \dots, \widehat{v}_j, \dots, v_p] \end{array}$$

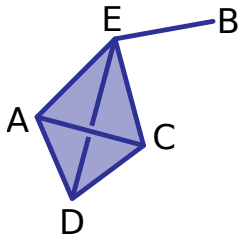
¹Warning: We are assuming $\mathbb{Z}_2$ coefficients from now on!

Checking linearity

$$\partial_p(\alpha + \beta) = \partial_p(\alpha) + \partial_p(\beta); \quad \alpha = \sum a_i \sigma_i, \quad \beta = \sum b_i \sigma_i$$

Homework

- $\partial_1([a, e]) =$



- $\partial_1([a, e] + [b, e]) =$

- $\partial_1([a, e] + [c, e] + [c, d] + [a, d]) =$

- $\partial_2([a, c, e] + [a, c, d]) =$